

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

## Unit – 5 : Session – 4 : SLO – 2 | Architecture Fundamentals + Amdahl's Law

### ■ I. Multiple Choice

1. c) Arithmetic Logic Unit (ALU) — executes the instruction's operation (the CU directs it).
2. b) RAM — volatile; loses contents without power.
3. c) Directs the flow of data and instructions.
4. a) Executing multiple instructions simultaneously in a sequence of stages.
5. b) Instruction-level parallelism (ILP).

### ■ II. True or False

1. TRUE.
2. FALSE — RAM is volatile; ROM/Flash are non-volatile.
3. TRUE.
4. TRUE.
5. FALSE — the CU controls; it does not store data permanently.

### ■ III. Scenario-Based Activity — Amdahl's Law

- Speedup with 8 cores:  $S = 1 / (0.30 + 0.70/8) = 1 / (0.30 + 0.0875) = 1 / 0.3875 \approx 2.58\times$ .
- Beyond 8 cores: no significant gains — the maximum theoretical speedup with infinitely many cores is  $1/0.30 \approx 3.33\times$ , so moving from  $8 \rightarrow \infty$  cores gains only  $\sim 0.75\times$  more; the 30% sequential portion dominates.
- Strategy for the sequential part: optimize that code itself (better algorithm, faster single-core path), or boost single-thread performance (higher clock, more cache, ILP); alternatively restructure the workload to shrink the serial fraction (e.g. parallelize parts of Task B, pipeline it with Task A).